AI Based Cardiovascular Risk Assessment System Using Explainable Machine Learning

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Index Terms - Cardiovascular mortality, clinical prediction, machine learning, calibration, explainability.

I. Introduction

A. Motivation and research scope

- Explain why a probability estimate needs reliable input data, calibration checks and understandable explanations.
- Define the target as cardiovascular death within 120 months after baseline examination. It includes fatal heart and cerebrovascular disease; it does not measure nonfatal events or diagnose existing disease.
- Position the application as a university research prototype using public linked data and synthetic demonstration cases [1].

B. Research questions and contributions

- RQ1: How can validation and preprocessing turn clinical and lifestyle records into reliable model inputs? Deliver a reproducible cohort and input-quality audit.
- RQ2: How well do fixed candidate models discriminate and calibrate the defined outcome? Deliver comparisons with event counts and uncertainty.
- RQ3: How can explanations and data-quality warnings improve interpretation? Deliver factor displays and a planned comprehension evaluation.

II. Related Work

A. Clinical prediction and probability quality

- Review conventional risk models and recent regression and tree methods. Compare population, endpoint, horizon, sample size, predictors and validation design.
- Discuss calibration alongside discrimination; a high ranking score alone cannot establish reliable probabilities [2].

B. Explainability and transparent reporting

- Review SHAP as model attribution, with attention to output scale and correlated predictors [3].
- Use TRIPOD+AI to organize transparent reporting of methods, uncertainty and limitations [4]. Complete the literature matrix as research continues.

III. Data and Study Population

A. Public sources and cohort construction

Describe the five NHANES survey cycles from 1999-2008 and their linkage to the CDC 2019 public-use mortality files. Baseline inputs come from 42 XPT files. Join by SEQN; retain the documented release with follow-up through December 2019 [1], [5]. Fig. 1 summarizes construction.

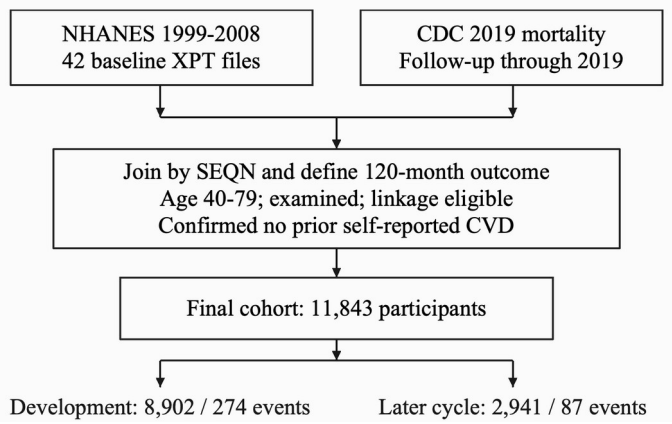


Fig. 1. Cohort construction and temporal split. Event counts refer to cardiovascular deaths within ten years. Source: project cohort audit [5].

- Include examined adults aged 40-79 with mortality-linkage eligibility and a known 120-month outcome.
- Require definite absence of prior self-reported heart failure, coronary disease, angina, myocardial infarction and stroke.
- Use mortality cause categories 001 and 005. Count known non-CVD deaths within the horizon as non-events; exclude indeterminate outcomes.

B. Eight baseline predictors

TABLE I
CURRENT MODEL INPUTS

Input	Unit or coding
Age	Years
Recorded male/female category	Two source categories
Mean systolic blood pressure	mmHg
Body mass index	kg/m²
Total cholesterol	mg/dL
HDL cholesterol	mg/dL
Current smoking	Yes / no
Self-reported diabetes	Yes / no

IV. Methodology

A. Design and separation of data

Describe the retrospective study and cycle-based split in Fig. 2. The 1999-2006 development sample has 8,902 participants and 274 events (3.08%). The 2007-2008 later cycle has 2,941 participants and 87 events (2.96%) [5].

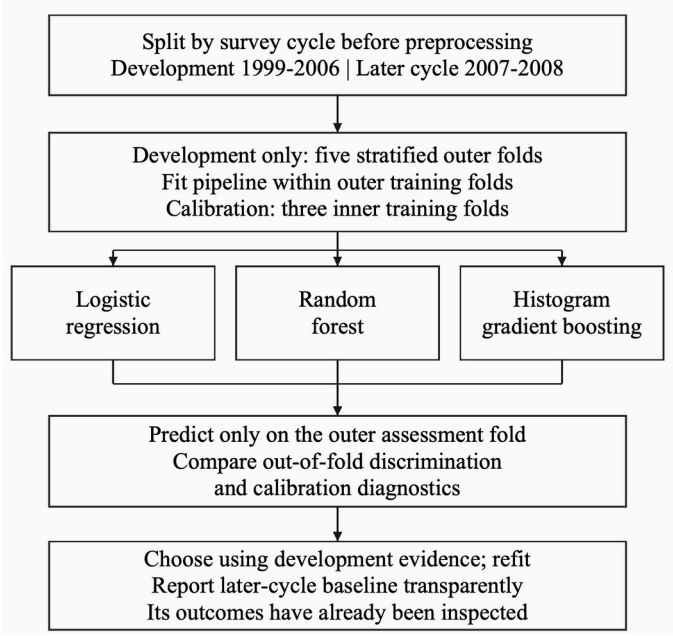


Fig. 2. Model-development and validation workflow. Preprocessing is fitted inside training folds. Calibration uses inner folds, while outer assessment predictions remain separate [5].

- Use identical five stratified folds for the fixed model comparison. No outcome or follow-up variable is a predictor.
- Disclose that later-cycle outcomes have already been inspected for the fixed logistic baseline. Further choices use development data only; this cycle is no longer a wholly unseen final test.

B. Preprocessing pipeline

- Check schema, units, source codes and missingness. Fit median imputation and standardization for continuous inputs within each training fold.
- Fit mode imputation and one-hot encoding for categorical inputs within the same fold. Preserve preprocessing with the serialized model.

C. Candidate models

- Logistic regression: L2 penalty, C = 1, no class weighting; the current web-model baseline.
- Random forest: 300 trees, maximum depth 6, minimum leaf size 20.
- Histogram gradient boosting: 200 iterations, learning rate 0.05, maximum 7 leaves, minimum leaf size 40, L2 = 1; not XGBoost or LightGBM.

D. Calibration experiment

- Compare original logistic output, sigmoid and isotonic calibration using five outer folds and three inner folds.
- Select with the written development-only rule: require supportive Brier improvement and calibration diagnostics. The present experiment retains the original output [5].

E. Performance and robustness

- Discrimination: AUROC and AUPRC, interpreted with sample size, event count and prevalence.
- Probability quality: Brier score, calibration plot, joint calibration intercept and slope [2].
- Precision: bootstrap intervals with their assumptions stated. Current person-level intervals hold predictions fixed and omit model-fitting and survey-design uncertainty.
- Planned checks: complete cases versus imputation, influential extremes, cycle/HDL sensitivity, and exploratory age and recorded-category subgroups.

F. Explanation and uncertainty methods

- Current factors are exact centered logistic contributions in log-odds. Planned SHAP work will specify background data, scale, numerical fidelity and stability [3].
- Keep missing-input warnings separate from model uncertainty. Explore bootstrap refitting without inventing a patient confidence score or interpreting attribution as causation.

V. System Implementation

Describe the existing React and FastAPI application in Fig. 3. Explain validation, saved-pipeline inference, the results display and the printable synthetic-case report [5].

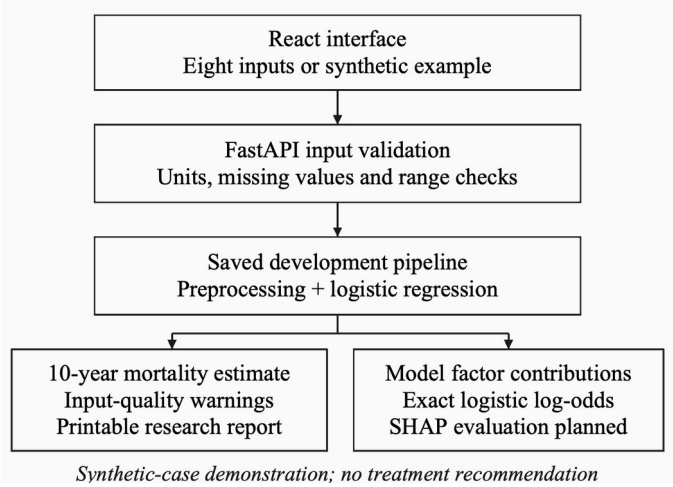


Fig. 3. Current research application architecture. Exact logistic attribution is implemented; SHAP validation is planned. No MRI processing is included.

VI. Preliminary Results

A. Development comparison completed

Use Table II to summarize the existing fixed comparison. Scores come from identical five-fold out-of-fold predictions for 8,902 participants, including 274 target events [5].

TABLE II
PRELIMINARY DEVELOPMENT PERFORMANCE

Model	AUROC	AUPRC	Brier
Logistic regression	0.777	0.124	0.02841
Random forest	0.776	0.119	0.02855
Histogram boosting	0.765	0.118	0.02866

- Logistic regression has the lowest Brier score in this comparison. Paired bootstrap intervals for differences include zero, so the small differences do not establish superiority.
- Calibration Brier scores: original 0.028412, sigmoid 0.028434, isotonic 0.028566. Neither adjustment met the written switching rule.

B. Later-cycle baseline completed

- For the fixed logistic baseline: AUROC 0.778 (preliminary 95% interval 0.738-0.819), AUPRC 0.080 and Brier 0.0281.
- Report 87 events among 2,941 participants. These unweighted sample metrics are not evidence of clinical transportability or diagnostic accuracy.

VII. Ongoing Research and Deliverables

A. Analyses still to complete

- Finalize the literature matrix and record sensitivity-analysis amendments before running new experiments.
- Complete missing-data, extreme-value and HDL/cycle checks. Specify survey-design-aware analysis using retained weights, strata and sampling units.
- Report subgroup denominators and precision; avoid strong fairness claims when event counts are small.

B. Explanation and application work

- Evaluate SHAP fidelity and stability. Refine accessible wording, missing-input messages and report presentation.
- Prepare a formative comprehension study using synthetic cases, subject to the institution's required review. Define tasks and measures before collecting responses.
- Package reproducible code, dependency versions, the data dictionary, model card and TRIPOD+AI report. Current checks establish software behavior, not clinical validity [4], [5].

VIII. Limitations and Ethical Considerations

A. Limits of the current evidence

- Discuss the historic US source population, self-reported history, missing inputs, HDL assay changes and limited later-cycle events.
- Disclose privacy substitutions in some public mortality follow-up times and causes [1]. Current unweighted scores omit the complex survey design.
- Separate future-risk estimation from diagnosis and treatment decisions. Explanations show model associations, not treatment effects.

B. Intended use and future extensions

- Use synthetic profiles for demonstrations. Obtain the applicable institutional determination for secondary-data research and review for any user study.
- Clinical use would require a defined intended use, independent target-population validation, privacy governance and authorized prospective evaluation.
- Treat cardiac MRI as a possible separate extension. Combined risk modeling would require paired scans, clinical inputs and suitable outcomes for the same people; unrelated MRI datasets cannot be joined to NHANES records.

References

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